

Verified theorem excerpt from Hyder, arXiv:1510.06662

This is a review transcription, not a facsimile of the answer paper. It was checked against the official 30-page arXiv PDF on 11 August 2026: [arXiv:1510.06662](https://arxiv.org/abs/1510.06662).

Theorem 1.1 (official PDF page 3). Let $\Omega \subset \mathbb{R}^n$ be open with finite measure, let $1 < p < \infty$, and let $f : [0, \infty) \rightarrow [0, \infty)$ be Borel measurable with $f(t) \rightarrow \infty$ as $t \rightarrow \infty$. With $p' = p/(p-1)$ and the sharp Martinazzi constant $\alpha_{n,p}$,

$$\sup_{\substack{u \in \tilde{H}^{n/p,p}(\Omega) \\ \|u\|_{L^p(\Omega)}^p + \|(-\Delta)^{n/(2p)}u\|_{L^p(\mathbb{R}^n)}^p \leq 1}} \int_{\Omega} f(|u|) e^{\alpha_{n,p}|u|^{p'}} dx = \infty.$$

The paper explicitly identifies this as a positive answer to Martinazzi's question and notes that the use of the full norm makes the result stronger.

Theorem 1.3 (official PDF page 5). Let $\Omega \subset \mathbb{R}^n$ be open with finite measure, $0 < \lambda < \lambda_1$, and $b > 0$. Then there is a nonzero weak solution of

$$(-\Delta)^{n/2}u = \lambda u e^{bu^2} \quad \text{in } \Omega, \quad u \in \tilde{H}^{n/2,2}(\Omega).$$

No parity restriction on n is imposed.

Bibliographic record. Ali Hyder, *Moser functions and fractional Moser–Trudinger type inequalities*, Nonlinear Analysis 146 (2016), 185–210, [doi:10.1016/j.na.2016.08.024](https://doi.org/10.1016/j.na.2016.08.024).